

Carson A Capps

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Education

Texas A&M University

May 2027

Master of Engineering in Aerospace Engineering (GPA: 4.00 / 4.00)

College Station, Texas

- **Relevant Coursework:** Modern Control of Aero Systems, Estimation of Dynamic Systems.

Texas A&M University

May 2026

Bachelor of Science in Aerospace Engineering (GPA: 3.59 / 4.00)

College Station, Texas

- **Relevant Coursework:** Applied Machine Learning for Aerospace Engineering, Spacecraft Attitude Dynamics and Control, Active Controls for Aerospace Vehicles.

Research Experience

Texas A&M University ASTRO Center

Jan 2025 – May 2026

Undergraduate Research Assistant

College Station, Texas

- Developed 6-DoF fuel optimal powered descent guidance for the lunar landing scenario using successive convexification (SCvx).
- Verified physics fidelity of SpaceTeamsPRO simulator descent simulations, confirming dynamical model accuracy for guidance V&V.
- Validated non-convex solver outputs against SpaceTeamsPRO physics engine to ensure trajectory solutions were dynamically realizable.
- Trained and validated transformer-based neural network to accelerate successive convexification solve time by up to 75% (T-SCvx), benchmarking solution accuracy against baseline solver.
- Conducted Monte Carlo analysis over terminal descent initial conditions derived using lunar deorbit dispersions from NASA's GMAT, characterizing T-SCvx robustness to realistic mission dispersion – IEEE Aerospace 2026.

Projects

Orbital Servicing Satellite Senior Design | ADCS Team Lead

- Led ADCS sub-team for autonomous LEO servicing spacecraft, building and integrating Basilisk simulation environment for 4-phase (sun-pointing, hill-pointing, rendezvous, and proximity operations) attitude V&V.
- Built Kalman filter using datasheet-specified noise and performance parameters of navigation sensors, confirming design satisfied navigation state accuracy requirements.

HALO-D: Vision-Based Autonomous Docking | Project Lead

- Developed a Decision Transformer guidance policy for autonomous docking, demonstrating vision-only RGB-depth image sequences as sufficient state representation for closed-loop docking guidance.
- Achieved >98% docking success when simulated in SpaceTeamsPRO simulation software, matching LQR expert.
- Implemented an SVD-based runtime anomaly detector and validated against synthetic fault classes.

Spacecraft Linear Control Design | Class Project

- Given open-loop behavior, designed PID and lead/lag compensators in MATLAB to meet spacecraft attitude control requirements including settling time, overshoot, phase margin, and gain margin specifications.
- Characterized closed-loop reference tracking, disturbance rejection, and noise attenuation requirements using MATLAB.

Fault Tolerant Ascent Guidance | Personal Project

- Designed convex-QP thrust allocation algorithm under Powered Explicit Guidance (PEG) for a 33-engine booster to reallocate thrust and maintain trajectory following simulated engine-out faults.
- Validated fault-tolerance and guidance robustness over a 100-flight Monte Carlo campaign spanning randomized engine-out fault cases.

Publications

W. Lee, C. Capps, J. Goldhammer, G. Chamitoff, "Mission Specific Trajectory Dispersions for T-SCvx Landing Guidance", in IEEE Aerospace, Mar 2026

Technical Skills

Languages: Python, C++, MATLAB

Tools: Git, PyTorch, NASA GMAT, SolidWorks, ANSYS

Concepts: GNC, Trajectory Optimization, Machine Learning, ADCS, Simulation and V&V